Context-Aware MTN for Pedestrian Trajectory Prediction

A CPU-based Proof-of-Concept on IDD-PeD

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Abstract

This draft documents a context-aware extension of MTN for pedestrian trajectory prediction on IDD-PeD. The proposed proof-of-concept architecture, C4 Context-Aware MTN, augments observed pedestrian trajectories with six native context sources and shares only encoder self-attention projections across depth. The draft reports a matched CPU Farneback evaluation and explicitly distinguishes it from the authors' RAFT-based benchmark. Dynamic Multimodal Fusion is included as a separate evaluated candidate rather than part of the proposed C4 architecture.

1 Proposed Methodology

1.1 Problem formulation and MTN backbone

Given an observed pedestrian bounding-box trajectory $\mathbf{X}_{obs} \in \mathbb{R}^{B\times 15\times 4}$, the objective is to predict future bounding boxes $\widehat{\mathbf{X}}_{fut} \in \mathbb{R}^{B\times 45\times 4}$. The MTN backbone embeds the observed trajectory and adds positional encoding, producing a sequence of width 32, i.e., $[B, 15, 32]$. Three MTN encoder layers generate trajectory memory $\mathbf{M} \in \mathbb{R}^{B\times 15\times 32}$. The original flow-related decoder paths are retained; the decoder output is mapped to four bounding-box coordinates and combined with the final observed-box residual. The dataset protocol and baseline implementation materials follow the official IDD-PeD release [1, 2].

1.2 Native IDD-PeD context branch

The first contribution is a context branch built only from six verified native IDD-PeD sources: *intersection_type*, *signalized_type*, *road_type*, *location_type*, *motion_direction*, and observed *occlusion*. Day/night and illumination are not included because they were not verified as inputs of the implemented final context branch.

Each source is represented by a 16-dimensional embedding. The five static labels use independent embedding lookups; the observed occlusion history is embedded and mean-pooled over observed steps. The six vectors are concatenated to a 96-dimensional context feature, then projected to the MTN width of 32. The original context-admission gate receives the context representation together with the final observed bounding box, and yields a gated context token $\mathbf{c} \in \mathbb{R}^{B\times 1\times 32}$. The token is appended *after* the trajectory encoder, yielding augmented decoder memory $[B, 16, 32]$. This preserves the baseline trajectory memory while allowing the decoder's existing source-attention path to condition on scene context.

1.3 C4: cross-layer encoder self-attention weight sharing

The second contribution is selective cross-layer sharing in the trajectory encoder. C4 aliases only the Q/K/V/O projections of `EncoderLayer.self_attn` across encoder layers 0–2. The encoder's `src_attn`, `cross_attn`, feed-forward networks, normalisation layers, decoder modules, context branch, and output head remain independent. Therefore, C4 is not full layer sharing.

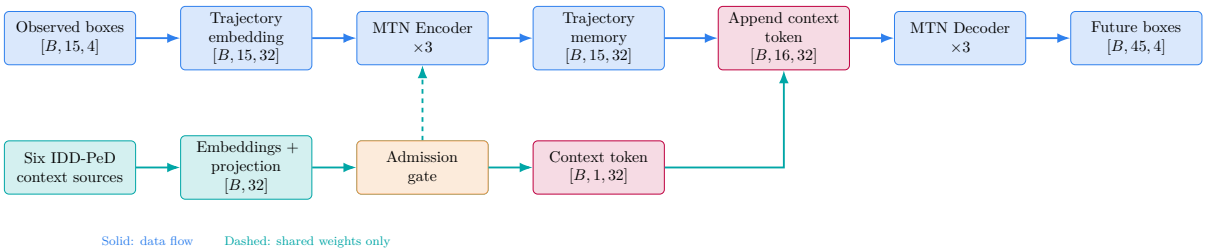


Figure 1: Colour-coded overview of the proposed C4 Context-Aware MTN. Blue: trajectory pathway; teal: native context pathway; orange: admission decision; purple: memory augmentation. Dashed teal denotes shared weights only.

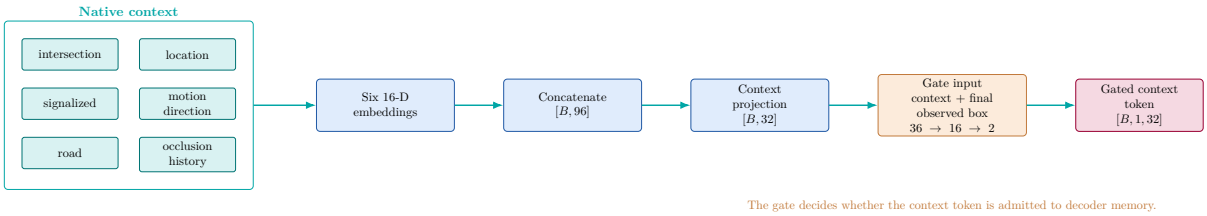


Figure 2: Context-token generation and admission gate. Each native context source is embedded before projection to the MTN width; the gate receives global context and the final observed box.

This choice reduces the number of unique parameters while preserving the three sequential encoder-layer executions. It must consequently be interpreted as a parameter-footprint technique, not as a FLOP-reduction claim. The measured CPU latency is reported separately in Section 2.

1.4 Novel contributions

C1. Context-token augmentation of MTN. The method introduces six verified IDD-PeD context sources as a gated 32-dimensional decoder-memory token without replacing the trajectory pathway.

C2. Selective, technically scoped encoder sharing. C4 ties only encoder self-attention Q/K/V/O across depth. This avoids incorrectly claiming that all encoder/decoder components are shared or that FLOPs decrease.

C3. Matched evidence-driven architecture selection. B, C4, F, and F+C4 were evaluated under a common small CPU protocol, enabling a factual comparison of accuracy, variation across seeds, parameter count, and latency.

1.5 Separate Dynamic Multimodal Fusion candidate

Dynamic Multimodal Fusion (F) was implemented and evaluated separately. It uses six 32-dimensional context tokens for cross-attention and a separate global context vector. The trajectory memory acts as query, while the six context tokens are keys and values. Identity, additive, and cross-attention candidates are computed, then a $64 \rightarrow 32 \rightarrow 3$ gate selects one operation per sample using hard Gumbel-Softmax in training and argmax/one-hot selection in evaluation. All three candidates are currently computed before selection; therefore F has no branch-skipping or resource-aware computational-saving claim. F is not included in the C4 proposed architecture in Figure 1.

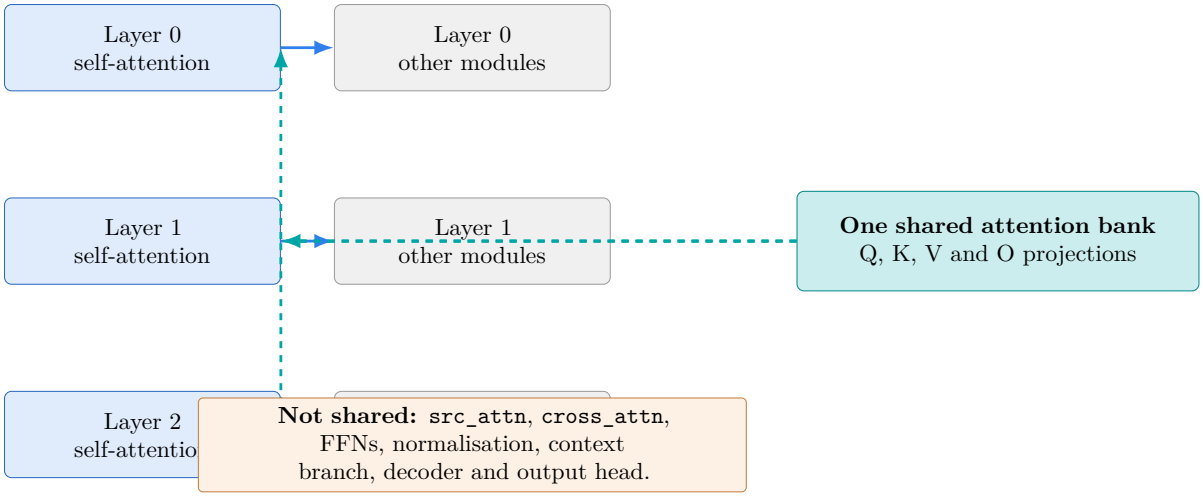


Figure 3: C4 selective cross-layer sharing. Only encoder self-attention Q/K/V/O projections are tied across depth; remaining modules are independent.

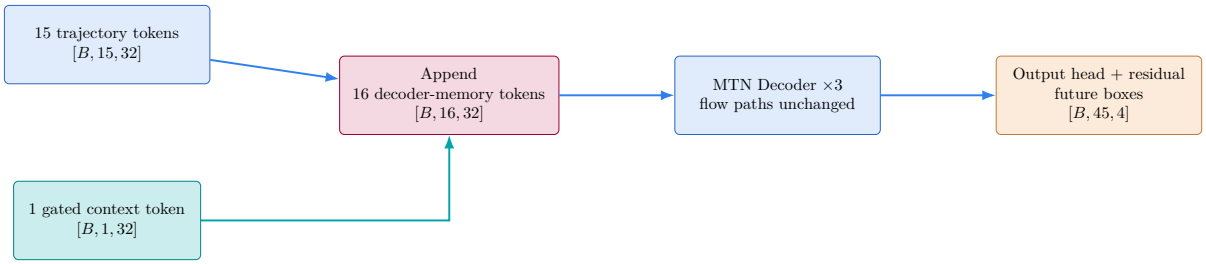


Figure 4: Decoder-memory augmentation. The gated context becomes the sixteenth decoder-memory token without replacing the original 15 trajectory tokens.

2 Experimental Results

2.1 Protocol and reporting boundary

The completed unified experiment used a fixed 32/8/24 train/validation/test trajectory split from locally available IDD-PeD data, CPU Farneback optical flow, Adam with learning rate 0.001, batch size 8, MSE loss, one training epoch, and three random seeds (20260928, 20260929, and 20260930). The test set comprises 24 trajectories from one held-out video. Reported values are mean ± sample standard deviation across the three seeds. This is a CPU-based proof-of-concept, not an official RAFT-based IDD-PeD benchmark reproduction.

2.2 Discussion of matched results

C4 achieved the lowest mean MSE, C-MSE, and CF-MSE in this matched POC. Relative to B, C4 changed MSE by −5.01%, C-MSE by −4.84%, and CF-MSE by −2.82%, while reducing unique parameters by 6.07%. Its mean CPU latency was 3.93% higher than B. The effect is not uniformly dominant per seed: C4 attained lower MSE than B in two of three seeds, while its MSE standard deviation was substantially larger (93.33 versus 20.16). Thus, the evidence supports C4 as the current proposed POC architecture, not a final universal selection.

F did not improve the measured prediction metrics over B: its relative changes were +1.55% MSE, +1.52% C-MSE, and +4.27% CF-MSE, accompanied by +10.68% parameters and +11.18% CPU latency. F+C4 reduced mean MSE by 0.92% relative to B but increased CF-MSE by 4.72%, parameters by 4.61%, and latency by 14.23%. These results do not justify including F or F+C4 in the current proposed C4 architecture.

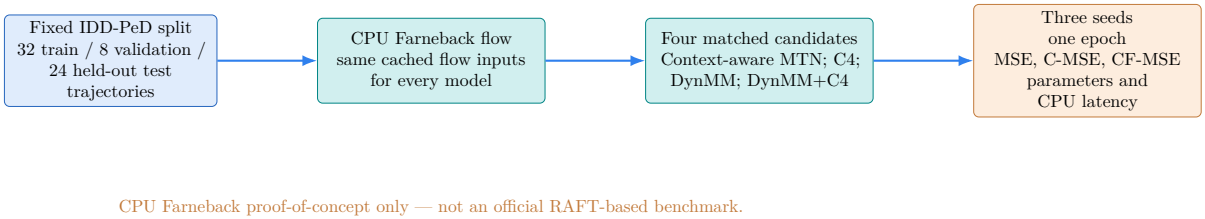


Figure 5: Matched CPU proof-of-concept protocol used for the completed unified experiment.

Table 1: Unified matched POC results. Lower MSE, C-MSE, CF-MSE, parameter count, and latency are better.

Model	MSE	C-MSE	CF-MSE	Parameters	CPU latency (ms)
B	1137.11 ± 20.16	1107.40 ± 19.63	2971.36 ± 103.13	139,142	45.58 ± 3.23
C4	$\mathbf{1080.16 \pm 93.33}$	$\mathbf{1053.83 \pm 95.13}$	$\mathbf{2887.51 \pm 237.14}$	**130,694**	47.38 ± 2.48
F	1154.78 ± 9.95	1124.20 ± 10.08	3098.16 ± 137.74	154,007	50.68 ± 2.48
F+C4	1126.67 ± 30.84	1098.43 ± 27.25	3111.62 ± 85.57	145,559	52.07 ± 4.74

2.3 Comparison with prior work on IDD-PeD

Table 2 reports the published 1.5-second IDD-PeD trajectory-baseline results from the official paper [1]. The published MTN, BiTraP, and SGNet results use the authors' approximately 70/30 train/test split, RAFT optical flow for MTN, GPU hardware, and the full test set. In contrast, our completed comparison uses CPU Farneback flow, one held-out video, 24 trajectories, one epoch, and three seeds. Therefore, the table is a *dataset-context comparison*, not a direct numerical ranking of our method against the published baselines.

The current POC has lower numerical error than the published deterministic MTN baseline, but this must not be interpreted as a benchmark improvement because flow representation, train/test scale, training budget, and hardware differ. The appropriate current conclusion is limited: within the matched local POC, encoder self-attention sharing improved mean error metrics relative to the local Context-Aware MTN reference. A direct comparison against published work requires reproducing the official split and RAFT pipeline.

3 Ablation Study

The ablation study is organised around completed experiments already documented in the project tracker. Results from different POC protocols are not merged into a single ranking; the unified 12-run ablation is the primary matched comparison.

3.1 Context contribution: original MTN versus B

An early paired POC compared original MTN (A) with context-aware MTN (B) on 10 held-out trajectories from GP6 video 3, using one seed, 12 epochs, and CPU Farneback flow. A obtained MSE 1242.98, C-MSE 1225.17, CF-MSE 4093.85, and latency 69.24 ms. B obtained MSE 1113.21, C-MSE 1094.57, CF-MSE 3646.51, and latency 61.41 ms. This preliminary result motivated retaining the context branch. It is not directly comparable with the later unified one-epoch, three-seed experiment.

3.2 Full sharing versus targeted sharing

Full encoder-and-decoder sharing reduced parameters from 139,142 to 53,894 (61.3%) but increased MSE from 1113.21 to 1339.31 and increased latency from 47.20 to 50.13 ms in its

Table 2: Published IDD-PeD baselines at 1.5 seconds and the current CPU POC. Lower is better. The final row is not directly comparable with the published rows because the protocol differs.

Source / protocol	Method	MSE	C-MSE	CF-MSE
Official IDD-PeD paper; full test split; RAFT/GPU	PIETraj	2181	1979	8960
Official IDD-PeD paper; full test split; RAFT/GPU	MTN	1652	1489	6668
Official IDD-PeD paper; full test split; published setup	BiTraP	338	200	638
Official IDD-PeD paper; full test split; published setup	SGNet	310	190	531
Current work; CPU Farneback POC, 24 trajectories, one epoch	Context-Aware MTN with Encoder Self-Attention Sharing	1080.16 ± 93.33	1053.83 ± 95.13	2887.51 ± 237.14

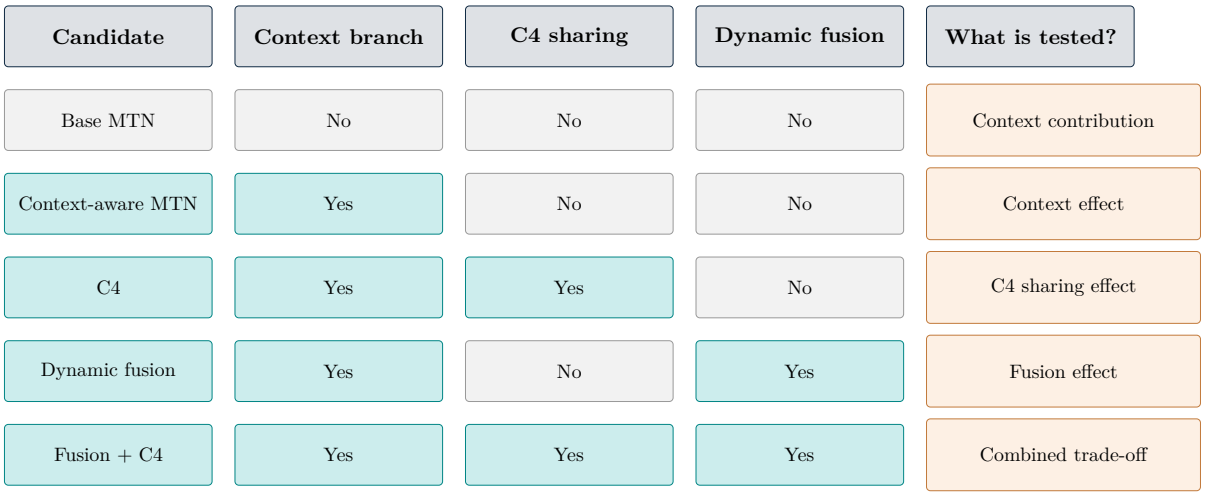


Figure 6: Ablation design: each candidate isolates the contribution of native context, C4 sharing, dynamic fusion, or their combination.

recorded POC. It was therefore rejected. This supports testing selective sharing rather than tying entire layers.

3.3 Unified architecture ablation

The primary ablation is Table 1, which compares B, C4, F, and F+C4 under the same split, seed set, optimiser, loss, epoch budget, and CPU flow cache. It separately tests: (i) C4 encoder self-attention sharing, (ii) F dynamic multimodal fusion, and (iii) their combination. This design demonstrates component contributions without changing the evaluation protocol between compared variants.

3.4 Interpretation and remaining ablations

The evidence indicates that selective C4 sharing is more promising than full sharing, decoder-targeted sharing, or the currently implemented F fusion. However, the observed C4 variance and latency increase require broader validation. The next ablation, if approved, should retain the same architecture and evaluate across additional held-out videos and a reproduced RAFT/GPU setting. It should measure the same MSE, C-MSE, CF-MSE, parameter count, latency, and seed-level variation. No additional result is claimed in this draft.

Table 3: Targeted weight-sharing POC. C3 was not run because the current decoder has no active decoder self-attention. C5 was not run because genuine head sharing requires a new projection parameterisation.

Variant	Sharing scope	Parameters	MSE	C-MSE	CF-MSE / latency (ms)
B	None	139,142	972.51	949.45	2802.67 / 47.79
C1	Decoder src_attn Q/K/V/O	130,694	1081.44	1049.76	3369.58 / 53.68
C2a	Decoder src_attn Q/K/V	132,806	1188.92	1169.76	3940.10 / 56.30
C2b	Decoder src_attn K/V	134,918	1152.49	1133.15	3807.02 / 47.99
C4	Encoder self_attn Q/K/V/O	130,694	819.99	798.82	2441.16 / 49.56

4 Limitations and Next Steps

The draft has four key limitations. First, CPU Farneback flow differs from the authors' RAFT pipeline. Second, the unified POC uses one held-out video and 24 trajectories. Third, models were trained for one epoch, so the comparison is not a convergence study. Fourth, C4 reduces unique parameters but does not reduce FLOPs because all encoder attention operations execute; F computes all fusion candidates before selecting one.

The immediate next steps are supervisor review of the C4 and F diagrams, integration of the selected C4 figure and text into the manuscript, and broader validation only if requested. The published values currently shown in Table 2 should remain labelled as a dataset-context comparison; they should be used for direct numerical ranking only after the official split, flow pipeline, training budget, and evaluation hardware are reproduced.

References

[1] Ruthvik Bokkasam, Shankar Gangisetty, A. H. Abdul Hafez, and C. V. Jawahar. Pedestrian intention and trajectory prediction in unstructured traffic using idd-ped. In *Proceedings of the IEEE International Conference on Robotics and Automation*, 2025.

[2] Ruthvik9. IDD-PeD: Official repository. GitHub repository. Accessed for implementation and reproduction materials.